B6 Condensed-Matter Physics — Model Solutions, Trinity 2021

Question 1 — Powder neutron diffraction of cubic $RbMnF_3$

Problem. Cubic $RbMnF_3$ powder, neutrons of $E = 25\,\mathrm{meV}$. Rings at $2\theta = 24.6°, 35.1°, 43.4°, 50.5°$. Find d_{hkl}, Miller indices, lattice type and accurate a ($\approx 0.4\,\mathrm{nm}$). Below $T_c = 83\,\mathrm{K}$ two new rings; one at 21.3°. Mechanism, predict the second.

(a) Why neutrons rather than X-rays

Neutrons carry a magnetic moment that couples to unpaired electron spin density; X-rays scatter from charge only. $\boxed{\text{Use neutrons to detect magnetic order (e.g. AFM } Mn^{2+} \text{ structure in } RbMnF_3\text{); X-rays cannot.}}$

(b) Lattice-plane spacings

Start from $E = p^2/(2m_n)$:

$$p = \sqrt{2m_n E}.$$

de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_n E}}.$$

Convert energy to joules:

$$E = 25 \times 10^{-3} \times 1.602 \times 10^{-19}\,\mathrm{J}.$$

Evaluate the product:

$$E = 4.005 \times 10^{-21}\,\mathrm{J}.$$

Plug into momentum formula:

$$p = \sqrt{2 \cdot 1.67493 \times 10^{-27} \cdot 4.005 \times 10^{-21}}\,\mathrm{kg\,m\,s^{-1}}.$$

Evaluate the radicand:

$$p = \sqrt{1.342 \times 10^{-47}}\,\mathrm{kg\,m\,s^{-1}}.$$

Take the square root:

$$p = 3.663 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}}.$$

Divide h by p:

$$\lambda = \frac{6.626 \times 10^{-34}}{3.663 \times 10^{-24}}\,\mathrm{m}.$$

Evaluate the quotient:

$$\lambda = 0.1809\,\mathrm{nm}.$$

Bragg's law:

$$d_{hkl} = \frac{\lambda}{2\sin\theta}, \qquad \theta = \tfrac{1}{2}(2\theta). \tag{1}$$

Tabulate:

2θ	θ	$\sin\theta$	d_{hkl}
$24.6°$	$12.30°$	0.2130	0.4247 nm
$35.1°$	$17.55°$	0.3015	0.3000 nm
$43.4°$	$21.70°$	0.3697	0.2447 nm
$50.5°$	$25.25°$	0.4266	0.2120 nm

$$\boxed{d_1 = 0.425\,\text{nm},\ d_2 = 0.300\,\text{nm},\ d_3 = 0.245\,\text{nm},\ d_4 = 0.212\,\text{nm}.} \tag{2}$$

(c) Miller indices, lattice type, and accurate a

Cubic spacing rule:

$$d_{hkl} \;=\; \frac{a}{\sqrt{N}}, \qquad N \equiv h^2 + k^2 + l^2. \tag{3}$$

Solve for N:

$$N \;=\; (a/d)^2.$$

Plug in $a \approx 0.4\,\text{nm}$ for each d:

$$N \;\approx\; \left(\tfrac{0.4}{0.4247}\right)^2,\ \left(\tfrac{0.4}{0.3000}\right)^2,\ \left(\tfrac{0.4}{0.2447}\right)^2,\ \left(\tfrac{0.4}{0.2120}\right)^2.$$

Evaluate each:

$$N \;\approx\; 0.887,\ 1.778,\ 2.671,\ 3.561.$$

Identify the integer ratios:

$$N \;=\; 1,\ 2,\ 3,\ 4.$$

Selection rules eliminate FCC and BCC: FCC requires hkl all-even or all-odd (first allowed $N = 3$), so $N = 1$ rules it out; BCC requires $h + k + l$ even (sequence $N = 2, 4, 6, 8, \ldots$), so $N = 1, 3$ rule it out. Hence $\boxed{\text{simple cubic.}}$

Miller indices in order: $(100), (110), (111), (200)$.

Invert spacing relation:

$$a \;=\; d\sqrt{N}.$$

Apply to ring 1 ($N = 1$):

$$a_1 \;=\; 0.4247 \cdot 1 \;=\; 0.4247\,\text{nm}.$$

Apply to ring 2 ($N = 2$):

$$a_2 \;=\; 0.3000 \cdot \sqrt{2} \;=\; 0.4243\,\text{nm}.$$

Apply to ring 3 ($N = 3$):

$$a_3 \;=\; 0.2447 \cdot \sqrt{3} \;=\; 0.4238\,\text{nm}.$$

Apply to ring 4 ($N = 4$):

$$a_4 \;=\; 0.2120 \cdot 2 \;=\; 0.4240\,\text{nm}.$$

Sum the four values:

$$\sum_i a_i \;=\; 0.4247 + 0.4243 + 0.4238 + 0.4240 \;=\; 1.6968\,\text{nm}.$$

Take the average:

$$\bar{a} \;=\; \tfrac{1}{4} \cdot 1.6968\,\text{nm}.$$

Evaluate the mean:

$$\boxed{a \;=\; 0.4242(4)\,\text{nm}.} \tag{4}$$

(d) Field changes ring intensity but not angle

Bragg angles depend only on d_{hkl}; field-induced lattice distortion (magnetostriction) is negligible, so positions are fixed. Neutron structure factor splits into nuclear and magnetic parts:

$$F_{hkl} = F_{hkl}^{\text{nuc}} + F_{hkl}^{\text{mag}}(\hat{\boldsymbol{B}}).$$

At room temperature ($T > T_c$) Mn^{2+} ions are paramagnetic; an applied $\boldsymbol{B}$ polarises the moments along $\hat{\boldsymbol{B}}$. Intensity depends on the component of magnetisation perpendicular to $\boldsymbol{Q}$:

$$|F^{\text{mag}}|^2 \propto |\hat{\boldsymbol{Q}}\times(\boldsymbol{M}\times\hat{\boldsymbol{Q}})|^2.$$

Hence $|F_{hkl}|^2$ varies with $\boldsymbol{B}$ without shifting the ring. $\boxed{\text{Field aligns Mn}^{2+}\text{ moments; magnetic structure factor chang}}$

(e) New rings below T_c: G-type antiferromagnetism

Mechanism: nearest-neighbour Mn moments alternate in sign on the simple-cubic Mn sublattice (G-type AFM). Magnetic unit cell is $2a \times 2a \times 2a$.

Magnetic reflections sit at $\boldsymbol{Q} = (2\pi/a)(h', k', l')$ with h', k', l' half-odd-integers; index

$$N_{\text{mag}} = h'^2 + k'^2 + l'^2.$$

Take the smallest set $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$:

$$N_{\text{mag}} = \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{3}{4}.$$

Magnetic plane spacing:

$$d^* = \frac{a}{\sqrt{N_{\text{mag}}}} = \frac{0.4242}{\sqrt{3/4}}\,\text{nm}.$$

Evaluate the quotient:

$$d^* = 0.4898\,\text{nm}.$$

Apply Bragg's law:

$$\sin\theta = \frac{\lambda}{2d^*} = \frac{0.1809}{2 \cdot 0.4898}.$$

Evaluate the ratio:

$$\sin\theta = 0.1847.$$

Take the arcsine:

$$\theta = 10.64^\circ.$$

Double to get 2θ:

$$2\theta = 21.3^\circ,$$

matching the observation. $\boxed{\text{G-type AFM ordering; magnetic period } 2a.}$

Next allowed magnetic reflection $(\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$:

$$N'_{\text{mag}} = \tfrac{9}{4} + \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{11}{4}.$$

Magnetic plane spacing:

$$d^{**} = \frac{a}{\sqrt{11/4}} = \frac{2 \cdot 0.4242}{\sqrt{11}}\,\text{nm}.$$

Evaluate the quotient:

$$d^{**} = 0.2558\,\text{nm}.$$

Apply Bragg's law:

$$\sin\theta \;=\; \frac{0.1809}{2\cdot 0.2558}.$$

Evaluate the ratio:

$$\sin\theta \;=\; 0.3537.$$

Take the arcsine:

$$\theta \;=\; 20.71^\circ.$$

Double to get 2θ:

$$\boxed{2\theta_{\text{2nd}} \;\approx\; 41.4^\circ.} \tag{5}$$

Slots between (110) at 35.1° and (111) at 43.4°, consistent with the observation window.

Question 2 — Crystal structure and phonons of AlP / Si

Problem. AlP plan view (zinc-blende). (a) Bravais lattice, lattice points, basis. (b) Reciprocal-lattice formula and derive the AlP reciprocal lattice. (c) Acoustic/optical phonon counts; longitudinal/transverse along [100] and number of distinct acoustic velocities. (d) For Si ($a = 0.5431\,\text{nm}$, $v_L = 8433\,\text{m}\,\text{s}^{-1}$ along [100]) estimate $\hbar\omega_{\text{LA}}$ at $\boldsymbol{k} = (2\pi/a, 0, 0)$. (e) Phonon-spectrum difference between AlP and Si at this $\boldsymbol{k}$.

(a) Bravais lattice and basis

Al sites at corners and all six face centres: FCC. P sites at $(\frac{1}{4},\frac{1}{4},\frac{1}{4})a$ and the three FCC translates: same FCC lattice shifted by $(\frac{1}{4},\frac{1}{4},\frac{1}{4})a$. This is the zinc-blende structure.

Conventional-cell FCC lattice points:

$$(0,0,0),\ (\tfrac{1}{2},\tfrac{1}{2},0),\ (\tfrac{1}{2},0,\tfrac{1}{2}),\ (0,\tfrac{1}{2},\tfrac{1}{2}).$$

$$\boxed{\text{Bravais: FCC. Basis: Al at } (0,0,0),\ \text{P at } (\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{4})a.} \tag{6}$$

(b) Reciprocal lattice

General formula ($\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}$):

$$\boldsymbol{b}_1 \;=\; 2\pi\,\frac{\boldsymbol{a}_2\times\boldsymbol{a}_3}{\boldsymbol{a}_1\cdot(\boldsymbol{a}_2\times\boldsymbol{a}_3)},\quad \boldsymbol{b}_2 \;=\; 2\pi\,\frac{\boldsymbol{a}_3\times\boldsymbol{a}_1}{\boldsymbol{a}_1\cdot(\boldsymbol{a}_2\times\boldsymbol{a}_3)},\quad \boldsymbol{b}_3 \;=\; 2\pi\,\frac{\boldsymbol{a}_1\times\boldsymbol{a}_2}{\boldsymbol{a}_1\cdot(\boldsymbol{a}_2\times\boldsymbol{a}_3)}. \tag{7}$$

FCC primitive vectors:

$$\boldsymbol{a}_1 \;=\; \tfrac{a}{2}(\hat{\boldsymbol{y}}+\hat{\boldsymbol{z}}),\quad \boldsymbol{a}_2 \;=\; \tfrac{a}{2}(\hat{\boldsymbol{z}}+\hat{\boldsymbol{x}}),\quad \boldsymbol{a}_3 \;=\; \tfrac{a}{2}(\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}).$$

Expand the cross product:

$$\boldsymbol{a}_2\times\boldsymbol{a}_3 \;=\; \tfrac{a^2}{4}(\hat{\boldsymbol{z}}+\hat{\boldsymbol{x}})\times(\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}).$$

Distribute and drop $\hat{\boldsymbol{x}}\times\hat{\boldsymbol{x}} = 0$:

$$\boldsymbol{a}_2\times\boldsymbol{a}_3 \;=\; \tfrac{a^2}{4}\big(\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}} + \hat{\boldsymbol{z}}\times\hat{\boldsymbol{y}} + \hat{\boldsymbol{x}}\times\hat{\boldsymbol{y}}\big).$$

Use $\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}\times\hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{x}}\times\hat{\boldsymbol{y}} = \hat{\boldsymbol{z}}$:

$$\boldsymbol{a}_2\times\boldsymbol{a}_3 \;=\; \tfrac{a^2}{4}(-\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}+\hat{\boldsymbol{z}}).$$

Take the triple product:

$$\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3) \;=\; \tfrac{a^3}{8}(\hat{\boldsymbol{y}} + \hat{\boldsymbol{z}}) \cdot (-\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}}).$$

Evaluate the dot product:

$$\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3) \;=\; \tfrac{a^3}{8}(0+1+0+0+0+1) \;=\; \tfrac{a^3}{4}.$$

Substitute into (**??**) for $\boldsymbol{b}_1$:

$$\boldsymbol{b}_1 \;=\; 2\pi\,\frac{(a^2/4)(-\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}})}{a^3/4}.$$

Cancel $a^2/4$ and simplify:

$$\boldsymbol{b}_1 \;=\; \frac{2\pi}{a}(-\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}}).$$

By cyclic symmetry:

$$\boldsymbol{b}_2 \;=\; \frac{2\pi}{a}(\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}}), \qquad \boldsymbol{b}_3 \;=\; \frac{2\pi}{a}(\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} - \hat{\boldsymbol{z}}).$$

Form pairwise sums to identify the conventional cubic axes:

$$\boldsymbol{b}_1 + \boldsymbol{b}_2 \;=\; \frac{4\pi}{a}\hat{\boldsymbol{z}}, \qquad \boldsymbol{b}_2 + \boldsymbol{b}_3 \;=\; \frac{4\pi}{a}\hat{\boldsymbol{x}}, \qquad \boldsymbol{b}_1 + \boldsymbol{b}_3 \;=\; \frac{4\pi}{a}\hat{\boldsymbol{y}}.$$

Recognise BCC primitives along cubic axes of edge $4\pi/a$.

$$\boxed{\text{Reciprocal lattice: BCC, conventional cubic edge } 4\pi/a.} \tag{8}$$

(c) Phonon-mode counting along [100]

Two atoms in the basis $\Rightarrow$ $3\times 2 = 6$ branches per $\boldsymbol{k}$: 3 acoustic, $3(2-1) = 3$ optical. $\boxed{\text{3 acoustic + 3 optical.}}$

Along [100], C_{4v} symmetry: 1 LA + 2 TA. The two TA modes are related by the four-fold rotation about [100], hence degenerate. $\boxed{\text{1 LA, 2 (degenerate) TA} \Rightarrow \text{two distinct acoustic velocities, } v_{\mathrm{LA}} \neq v_{\mathrm{TA}}.}$

(d) LA phonon energy at $\boldsymbol{k} = (2\pi/a, 0, 0)$ in Si

Approximate Si along [100] by a 1D monatomic chain with spacing $a_{\mathrm{ch}} = a/2$ so the BZ boundary $\pi/a_{\mathrm{ch}} = 2\pi/a$ coincides with X.

Monatomic-chain dispersion:

$$\omega(k) \;=\; 2\sqrt{K/m}\,\big|\sin(ka_{\mathrm{ch}}/2)\big|.$$

Linearise at small k:

$$\omega \;\approx\; \sqrt{K/m}\,ka_{\mathrm{ch}}.$$

Identify with sound speed $\omega = v_L k$:

$$\sqrt{K/m}\,a_{\mathrm{ch}} \;=\; v_L.$$

Eliminate $\sqrt{K/m}$:

$$\omega(k) \;=\; \frac{2v_L}{a_{\mathrm{ch}}}\big|\sin(ka_{\mathrm{ch}}/2)\big|.$$

Substitute $a_{\mathrm{ch}} = a/2$:

$$\omega(k) \;=\; \frac{4v_L}{a}\big|\sin(ka/4)\big|.$$

Evaluate the argument at $k = 2\pi/a$:

$$ka/4 \;=\; \pi/2.$$

Evaluate ω:

$$\omega \;=\; \frac{4v_L}{a}\sin(\pi/2).$$

Reduce $\sin(\pi/2) = 1$:

$$\omega \;=\; \frac{4v_L}{a}.$$

Multiply by $\hbar$:

$$\hbar\omega \;=\; \frac{4\hbar v_L}{a}.$$

Plug in numbers:

$$\hbar\omega \;=\; \frac{4\cdot 1.0546\times 10^{-34}\cdot 8433}{0.5431\times 10^{-9}}\,\text{J}.$$

Evaluate the numerator:

$$4\cdot 1.0546\times 10^{-34}\cdot 8433 \;=\; 3.557\times 10^{-30}.$$

Divide by the denominator:

$$\hbar\omega \;=\; \frac{3.557\times 10^{-30}}{5.431\times 10^{-10}}\,\text{J} \;\approx\; 6.55\times 10^{-21}\,\text{J}.$$

Convert to electronvolts:

$$\hbar\omega \;=\; \frac{6.55\times 10^{-21}}{1.602\times 10^{-19}}\,\text{eV} \;\approx\; 0.041\,\text{eV}.$$

$$\boxed{\hbar\omega_{\text{LA}}(2\pi/a,0,0) \;\approx\; 40\,\text{meV}.} \tag{9}$$

Approximations: 1D chain replaces 3D dynamics; sinusoidal dispersion extrapolated from sound speed; harmonic. Experimental Si LA at X is $\sim 50\,\text{meV}$, agreement within a factor of ~ 1.2.

(e) Why AlP and Si differ at $\boldsymbol{k} = (2\pi/a, 0, 0)$

In Si the two basis atoms are identical: nonsymmorphic diamond symmetry forces LA(X) =LO(X) and TA(X) =TO(X) (degenerate at the zone boundary).

In AlP the two basis atoms differ in mass ($M_{\text{Al}} = 26.98$ u, $M_{\text{P}} = 30.97$ u). Solve the 1D diatomic-chain dispersion at $k = \pi/a_{\text{ch}}$:

$$\omega^2 \;=\; K\Big(\tfrac{1}{M_1}+\tfrac{1}{M_2}\Big) \pm K\Big|\tfrac{1}{M_1}-\tfrac{1}{M_2}\Big|.$$

Take the $-$ root (LA) with $M_{\max} > M_{\min}$:

$$\omega_{\text{LA}}^2 \;=\; \frac{2K}{M_{\max}}.$$

Take the $+$ root (LO):

$$\omega_{\text{LO}}^2 \;=\; \frac{2K}{M_{\min}}.$$

Subtract:

$$\omega_{\text{LO}}^2 - \omega_{\text{LA}}^2 \;=\; 2K\left(\frac{1}{M_{\min}} - \frac{1}{M_{\max}}\right) \;>\; 0.$$

LA mode: heavy P oscillates, Al fixed. LO mode: light Al oscillates, P fixed.

Additionally AlP is partly ionic, so long-range Coulomb dipole field gives extra LO–TO splitting (absent in covalent Si). $\boxed{\text{Si: LA}(X) = \text{LO}(X).\ \text{AlP: gap opens (LO above LA) from Al/P mass mismatch + ionicity}}$

Question 3 — 2D heat capacity: phonons, free electrons, gapless semiconductor

Problem. (a) 2D insulator: C_v/N_a at high T and trend on cooling. (b) 2D electron gas at $k_BT \gg E_F$: C_v/N_e and trend on cooling. (c)–(d) 2D undoped semiconductor with $E_{\text{gap}} \to 0$, $m_e^* = m_h^* \equiv m^*$: density of states; $N_c(T)$, C_v^{tot}/N_c; reconcile with equipartition. (e) Ratio C_v^{tot}/C_v when $k_BT \ll E_F$ in (b) but $k_BT \gg E_{\text{gap}}$ in (c)/(d). Useful: $\int_0^\infty \mathrm{d}x/(e^x+1) = \ln 2$, $\int_0^\infty x\,\mathrm{d}x/(e^x+1) = \pi^2/12$.

(a) 2D insulator

Each 2D atom: 2 kinetic + 2 potential quadratic degrees of freedom $\Rightarrow$ equipartition $4 \cdot \frac{1}{2}k_B = 2k_B$ per atom.

$$\boxed{(C_v/N_a)_{T\to\infty} \;=\; 2k_B.} \tag{10}$$

2D Debye: $C_v \propto T^2$ at low T. $\boxed{\text{Decreases on cooling.}}$

(b) 2D classical electron gas, $k_BT \gg E_F$

Equipartition (2 translational kinetic d.o.f.):

$$\boxed{(C_v/N_e)_{k_BT\gg E_F} \;=\; k_B.} \tag{11}$$

Sommerfeld at $k_BT \ll E_F$: $C_v \propto T \to 0$. $\boxed{\text{Decreases on cooling.}}$

(c) 2D density of states

Set band edges at $E = 0$ (gap $\to 0$). Parabolic 2D dispersions:

$$E_c(\boldsymbol{k}) \;=\; +\frac{\hbar^2k^2}{2m^*}, \qquad E_v(\boldsymbol{k}) \;=\; -\frac{\hbar^2k^2}{2m^*}.$$

k-space DOS (per area, two spins):

$$g(\boldsymbol{k})\,\mathrm{d}^2k \;=\; \frac{2}{(2\pi)^2}\,\mathrm{d}^2k.$$

Use circular shells $\mathrm{d}^2k = 2\pi k\,\mathrm{d}k$:

$$g(\boldsymbol{k})\,\mathrm{d}^2k \;=\; \frac{2}{(2\pi)^2}\cdot 2\pi k\,\mathrm{d}k \;=\; \frac{k}{\pi}\,\mathrm{d}k.$$

Use $E = \hbar^2k^2/(2m^*)$ to get $k\,\mathrm{d}k = (m^*/\hbar^2)\,\mathrm{d}E$:

$$g_c(E)\,\mathrm{d}E \;=\; \frac{1}{\pi}\cdot\frac{m^*}{\hbar^2}\,\mathrm{d}E \;=\; \frac{m^*}{\pi\hbar^2}\,\mathrm{d}E.$$

Read off the (constant) DOS:

$$\boxed{g_c(E) \;=\; g_v(E) \;=\; \frac{m^*}{\pi\hbar^2} \quad \text{(each band).}} \tag{12}$$

(d) $N_c(T)$, $C_v^{\mathbf{tot}}/N_c$, equipartition check

Particle–hole symmetry ($m_e^* = m_h^*$) pins $\mu = 0$ at all T.

Conduction-band electron density per area:

$$n_c \;=\; \int_0^\infty \mathrm{d}E\, g_c(E)\, \frac{1}{e^{E/k_BT}+1}.$$

Insert the constant DOS:

$$n_c \;=\; \frac{m^*}{\pi\hbar^2}\int_0^\infty \frac{\mathrm{d}E}{e^{E/k_BT}+1}.$$

Substitute $x = E/k_BT$:

$$n_c \;=\; \frac{m^* k_BT}{\pi\hbar^2}\int_0^\infty \frac{\mathrm{d}x}{e^x+1}.$$

Use $\int = \ln 2$:

$$n_c \;=\; \frac{m^* k_BT\,\ln 2}{\pi\hbar^2}.$$

Multiply by area A:

$$\boxed{N_c(T) \;=\; A\,\frac{m^* k_BT\,\ln 2}{\pi\hbar^2}.} \tag{13}$$

Conduction-band thermal energy per area:

$$u_c \;=\; \int_0^\infty \mathrm{d}E\, g_c\, E\, \frac{1}{e^{E/k_BT}+1}.$$

Insert the DOS:

$$u_c \;=\; \frac{m^*}{\pi\hbar^2}\int_0^\infty \frac{E\,\mathrm{d}E}{e^{E/k_BT}+1}.$$

Substitute $x = E/k_BT$, $\mathrm{d}E = k_BT\,\mathrm{d}x$:

$$u_c \;=\; \frac{m^*(k_BT)^2}{\pi\hbar^2}\int_0^\infty \frac{x\,\mathrm{d}x}{e^x+1}.$$

Use $\int = \pi^2/12$:

$$u_c \;=\; \frac{m^*(k_BT)^2}{\pi\hbar^2}\cdot\frac{\pi^2}{12}.$$

Cancel one π:

$$u_c \;=\; \frac{\pi m^* k_B^2 T^2}{12\hbar^2}.$$

By particle–hole symmetry the valence band contributes equally:

$$u_{\mathrm{tot}} \;=\; 2u_c \;=\; \frac{\pi m^* k_B^2 T^2}{6\hbar^2}.$$

Differentiate w.r.t. T:

$$\frac{C_v^{\mathrm{tot}}}{A} \;=\; \frac{\partial u_{\mathrm{tot}}}{\partial T} \;=\; \frac{\pi m^* k_B^2 T}{3\hbar^2}. \tag{14}$$

Divide by n_c from (**??**):

$$\frac{C_v^{\mathrm{tot}}}{N_c} \;=\; \frac{\pi m^* k_B^2 T/(3\hbar^2)}{m^* k_BT\,\ln 2/(\pi\hbar^2)}.$$

Cancel m^*, T, $\hbar^2$:

$$\frac{C_v^{\rm tot}}{N_c} = \frac{\pi\cdot\pi}{3\ln 2}\,k_B.$$

Reduce:

$$\boxed{\frac{C_v^{\rm tot}}{N_c} = \frac{\pi^2}{3\ln 2}\,k_B \approx 4.74\,k_B.} \tag{15}$$

No equipartition violation: equipartition assumes a fixed number of quadratic d.o.f., but here $N_c(T)\propto T$ grows with temperature. Heating creates new carriers (and matching holes), so part of $C_v^{\rm tot}$ goes into producing extra excitations, not just sharing energy among a fixed set. Dividing by N_c alone undercounts the active particles (electrons *plus* holes); the per-particle ratio still exceeds the rigid k_B because the carrier population itself is thermally generated.

(e) Ratio $C_v^{\rm tot}/C_v$

2D Fermi gas at $k_BT \ll E_F$ via Sommerfeld with constant DOS $g(E_F) = m/(\pi\hbar^2)$:

$$\frac{C_v}{A} = \frac{\pi^2}{3}\,g(E_F)\,k_B^2T.$$

Insert the DOS:

$$\frac{C_v}{A} = \frac{\pi^2}{3}\cdot\frac{m}{\pi\hbar^2}\,k_B^2T.$$

Cancel one π:

$$\frac{C_v}{A} = \frac{\pi m\,k_B^2T}{3\hbar^2}.$$

Form the ratio with $C_v^{\rm tot}/A$ from (**??**):

$$\frac{C_v^{\rm tot}}{C_v} = \frac{\pi m^* k_B^2T/(3\hbar^2)}{\pi m\,k_B^2T/(3\hbar^2)}.$$

Cancel π, k_B^2T, $\hbar^2$:

$$\boxed{\frac{C_v^{\rm tot}}{C_v} = \frac{m^*}{m}.} \tag{16}$$

Both regimes give heat capacity linear in T with the same numerical prefactor; the ratio is set purely by the ratio of effective DOS at μ in the two systems, i.e. by m^*/m.

Question 4 — Magnetism of GaAs and shallow donors

Problem. (a) Magnetic class of undoped GaAs at $T=0$. (b) Larmor–Langevin $\chi = -\mu_0\rho e^2\langle r^2\rangle/(6m_e)$ for undoped GaAs. (c) Light n-doping with Si: binding radius, binding energy, and the temperature at which donor spin and orbital susceptibilities cancel. (d) Below density $1/a_{\rm bind}^3$ the impurity-band width is $W = E_{\rm bind}\exp\big(-\alpha n^{-1/3}/a_{\rm bind}\big)$; explain the exponential. (e) At $T=0$, write $\chi_{\rm spin}^{\rm IB}(n)$ with $\alpha=1$; find the dopant density at which it equals $|\chi_{\rm LL}|$ from (b), and the upper temperature for validity. GaAs data: $\rho_m = 5.9\,{\rm g\,cm^{-3}}$, $E_g = 1.4\,{\rm eV}$, $m^* = 0.067m_e$, $g=-0.45$, $\epsilon_r = 12.9$.

(a) Magnetic class

Filled valence band, empty conduction band, no localised moments, no Pauli paramagnetism (zero DOS at E_F), no exchange. Only orbital response (Lenz): $\boxed{\text{Diamagnetic.}}$

(b) Larmor–Langevin χ for undoped GaAs

Formula-unit mass:

$$m_{\text{f.u.}} = (69.7 + 74.9)\,\text{u} = 144.6\,\text{u}.$$

Convert to kg:

$$m_{\text{f.u.}} = 144.6 \cdot 1.6605 \times 10^{-27}\,\text{kg} = 2.401 \times 10^{-25}\,\text{kg}.$$

Number density of formula units:

$$n_{\text{f.u.}} = \frac{\rho_m}{m_{\text{f.u.}}} = \frac{5.9 \times 10^3}{2.401 \times 10^{-25}}\,\text{m}^{-3}.$$

Evaluate the quotient:

$$n_{\text{f.u.}} = 2.46 \times 10^{28}\,\text{m}^{-3}.$$

Electrons per formula unit:

$$Z_{\text{tot}} = Z_{\text{Ga}} + Z_{\text{As}} = 31 + 33 = 64.$$

Electron number density:

$$\rho_e = Z_{\text{tot}}\, n_{\text{f.u.}} = 64 \cdot 2.46 \times 10^{28}\,\text{m}^{-3}.$$

Evaluate the product:

$$\rho_e \approx 1.57 \times 10^{30}\,\text{m}^{-3}.$$

Approximate $\langle r^2 \rangle \sim a_0^2$ (rough, $\sim 30\%$):

$$\langle r^2 \rangle \approx (0.529 \times 10^{-10})^2\,\text{m}^2.$$

Evaluate the square:

$$\langle r^2 \rangle \approx 2.80 \times 10^{-21}\,\text{m}^2.$$

Substitute into $\chi = -\mu_0 \rho_e e^2 \langle r^2 \rangle / (6 m_e)$:

$$\chi_{\text{LL}} = -(1.257 \times 10^{-6}) \cdot \frac{1.57 \times 10^{30} \cdot (1.602 \times 10^{-19})^2}{6 \cdot 9.109 \times 10^{-31}} \cdot 2.80 \times 10^{-21}.$$

Evaluate e^2:

$$e^2 = (1.602 \times 10^{-19})^2 = 2.566 \times 10^{-38}\,\text{C}^2.$$

Evaluate $6m_e$:

$$6m_e = 6 \cdot 9.109 \times 10^{-31} = 5.466 \times 10^{-30}\,\text{kg}.$$

Group the fraction:

$$\frac{\rho_e e^2}{6m_e} = \frac{1.57 \times 10^{30} \cdot 2.566 \times 10^{-38}}{5.466 \times 10^{-30}} \approx 7.37 \times 10^{21}\,\text{m}^{-3}\,\text{C}^2\,\text{kg}^{-1}.$$

Multiply by $\mu_0 \langle r^2 \rangle$:

$$\chi_{\text{LL}} \approx -(1.257 \times 10^{-6}) \cdot 7.37 \times 10^{21} \cdot 2.80 \times 10^{-21}.$$

Evaluate the product:

$$\chi_{\text{LL}} \approx -2.6 \times 10^{-5}.$$

Round to one significant figure:

$$\boxed{\chi_{\text{LL}} \sim -3 \times 10^{-5} \text{ (SI, dimensionless).}} \tag{17}$$

Order-of-magnitude consistent with tabulated values for covalent semiconductors.

(c) Donor binding radius, energy, and cancellation T^*

Use hydrogenic rescaling $e^2 \to e^2/\epsilon_r$, $m_e \to m^*$:

$$a_{\text{bind}} = \frac{4\pi\epsilon_0\epsilon_r\hbar^2}{m^*e^2}.$$

Factor out the Bohr radius:

$$a_{\text{bind}} = a_0 \frac{\epsilon_r m_e}{m^*}.$$

Plug in numbers:

$$a_{\text{bind}} = 0.529\,\text{Å} \cdot \frac{12.9}{0.067}.$$

Evaluate the ratio:

$$\frac{12.9}{0.067} = 192.5.$$

Multiply by a_0:

$$a_{\text{bind}} = 0.529 \cdot 192.5\,\text{Å} \approx 10.2\,\text{nm}.$$

Likewise rescale the Rydberg:

$$E_{\text{bind}} = \text{Ry} \cdot \frac{m^*/m_e}{\epsilon_r^2}.$$

Plug in numbers:

$$E_{\text{bind}} = 13.6\,\text{eV} \cdot \frac{0.067}{12.9^2}.$$

Evaluate ϵ_r^2:

$$\epsilon_r^2 = 12.9^2 = 166.4.$$

Evaluate the rescaling factor:

$$\frac{0.067}{166.4} = 4.027 \times 10^{-4}.$$

Multiply by the Rydberg:

$$E_{\text{bind}} = 13.6 \cdot 4.027 \times 10^{-4}\,\text{eV} \approx 5.5\,\text{meV}.$$

$$\boxed{a_{\text{bind}} \approx 10.2\,\text{nm}, \qquad E_{\text{bind}} \approx 5.5\,\text{meV}.} \tag{18}$$

Donor spin Curie law ($S = \frac{1}{2}$):

$$\chi_{\text{spin}} = \frac{\mu_0 n_d g^2 \mu_B^2}{4k_BT}.$$

Donor orbital Larmor with effective mass m^* and hydrogenic $\langle r^2 \rangle_{1s} = 3a_{\text{bind}}^2$:

$$\chi_{\text{orb}} = -\mu_0 \frac{n_d e^2}{6m^*} \cdot 3a_{\text{bind}}^2.$$

Simplify:

$$\chi_{\text{orb}} = -\frac{\mu_0 n_d e^2 a_{\text{bind}}^2}{2m^*}.$$

Impose cancellation $\chi_{\text{spin}} + \chi_{\text{orb}} = 0$:

$$\frac{\mu_0 n_d g^2 \mu_B^2}{4k_BT^*} = \frac{\mu_0 n_d e^2 a_{\text{bind}}^2}{2m^*}.$$

Cancel $\mu_0 n_d$ and solve for $k_B T^*$:

$$k_B T^* \;=\; \frac{g^2 \mu_B^2\, m^*}{2e^2 a_{\text{bind}}^2}.$$

Substitute $\mu_B^2 = e^2\hbar^2/(4m_e^2)$:

$$k_B T^* \;=\; \frac{g^2 e^2 \hbar^2 m^*}{8m_e^2\, e^2 a_{\text{bind}}^2}.$$

Cancel e^2:

$$k_B T^* \;=\; \frac{g^2 \hbar^2 m^*}{8m_e^2 a_{\text{bind}}^2}.$$

Use $E_{\text{bind}} = \hbar^2/(2m^* a_{\text{bind}}^2)$, hence $\hbar^2/a_{\text{bind}}^2 = 2m^* E_{\text{bind}}$:

$$k_B T^* \;=\; \frac{g^2 \cdot 2m^*\, E_{\text{bind}} \cdot m^*}{8m_e^2}.$$

Collect factors:

$$\boxed{k_B T^* \;=\; \frac{g^2}{4}\left(\frac{m^*}{m_e}\right)^2 E_{\text{bind}}.} \tag{19}$$

Plug in $g^2 = 0.2025$, $(m^*/m_e)^2 = 4.49 \times 10^{-3}$, $E_{\text{bind}} = 5.5\,\text{meV}$:

$$k_B T^* \;=\; \frac{0.2025 \cdot 4.49 \times 10^{-3} \cdot 5.5 \times 10^{-3}}{4}\,\text{eV}.$$

Evaluate the numerator:

$$0.2025 \cdot 4.49 \times 10^{-3} \cdot 5.5 \times 10^{-3} \;=\; 5.00 \times 10^{-6}.$$

Divide by 4:

$$k_B T^* \;\approx\; 1.25 \times 10^{-6}\,\text{eV}.$$

Convert via $k_B = 8.617 \times 10^{-5}\,\text{eV/K}$:

$$T^* \;=\; \frac{1.25 \times 10^{-6}}{8.617 \times 10^{-5}}\,\text{K}.$$

Evaluate the quotient:

$$\boxed{T^* \;\approx\; 15\,\text{mK}.} \tag{20}$$

(d) Why the impurity bandwidth is exponentially small

Donor 1s wavefunction decays as $e^{-r/a_{\text{bind}}}$. Tight-binding hopping t between donors at separation $r \sim n^{-1/3}$ is the wavefunction overlap, also $\propto e^{-r/a_{\text{bind}}}$. Bandwidth $W \sim t$, with prefactor E_{bind} (the only available energy scale):

$$W \;\sim\; E_{\text{bind}}\, \exp\!\Big(-\alpha n^{-1/3}/a_{\text{bind}}\Big)\,.$$

$\boxed{\text{Exponential = wavefunction overlap of neighbouring 1s donor states.}}$

(e) Impurity-band spin χ, crossover density, validity

For $k_BT \ll W$ the half-filled impurity band is a Pauli system. DOS $\sim n/W$ (one state per donor in band of width W):

$$\chi_{\rm spin}^{\rm IB} \;\sim\; \mu_0 (g\mu_B)^2\, \frac{n}{W}.$$

Substitute W with $\alpha = 1$:

$$\boxed{\chi_{\rm spin}^{\rm IB} \;\sim\; \mu_0\, \frac{g^2\mu_B^2 n}{E_{\rm bind}}\, \exp\!\left(\frac{1}{n^{1/3}\, a_{\rm bind}}\right).} \tag{21}$$

Crossover $\chi_{\rm spin}^{\rm IB} \sim |\chi_{\rm LL}| = 3\times10^{-5}$. Evaluate the prefactor at $n_0 \equiv 1/a_{\rm bind}^3 \approx 10^{24}\,{\rm m^{-3}}$:

$$\mu_0(g\mu_B)^2 \;=\; (1.257\times10^{-6})\cdot 0.2025\cdot(9.27\times10^{-24})^2.$$

Evaluate the product:

$$\mu_0(g\mu_B)^2 \;\approx\; 2.19\times10^{-53}\,{\rm J^2\,T^{-2}\,m^{-1}\,A^{-2}}.$$

Convert $E_{\rm bind}$ to joules:

$$E_{\rm bind} \;=\; 5.5\times10^{-3}\cdot 1.602\times10^{-19} \;\approx\; 8.81\times10^{-22}\,{\rm J}.$$

Divide:

$$\frac{\mu_0 g^2\mu_B^2}{E_{\rm bind}} \;\approx\; \frac{2.19\times10^{-53}}{8.81\times10^{-22}} \;\approx\; 2.5\times10^{-32}\,{\rm m^3}.$$

Multiply by $n_0 = 10^{24}\,{\rm m^{-3}}$:

$$\frac{\mu_0 g^2\mu_B^2 n_0}{E_{\rm bind}} \;\approx\; 2.5\times10^{-8}.$$

Match $|\chi_{\rm LL}| = 3\times10^{-5}$ requires exponential factor:

$$\exp\!\left(\frac{1}{n^{1/3}a_{\rm bind}}\right) \;\sim\; \frac{3\times10^{-5}}{2.5\times10^{-8}} \;\sim\; 10^3.$$

Take the logarithm:

$$\frac{1}{n^{1/3}a_{\rm bind}} \;\sim\; \ln\!\big(10^3\big) \;\approx\; 7.$$

Solve for n^*:

$$n^* \;\sim\; \frac{1}{(7a_{\rm bind})^3} \;\sim\; \frac{1}{343\, a_{\rm bind}^3}.$$

Evaluate using $a_{\rm bind}^3 \approx 10^{-24}\,{\rm m^3}$:

$$n^* \;\sim\; 3\times10^{21}\,{\rm m^{-3}} \;\sim\; 3\times10^{15}\,{\rm cm^{-3}}.$$

$$\boxed{n^* \;\sim\; 10^{15}\text{–}10^{16}\,{\rm cm^{-3}} \quad \text{(well below the Mott density } \sim 10^{18}\,{\rm cm^{-3}}\text{).}} \tag{22}$$

Validity: Pauli regime requires $k_BT \ll W$. At n^*:

$$W \;\sim\; E_{\rm bind}\, e^{-7} \;\approx\; 5.5\,{\rm meV}\cdot 9\times10^{-4} \;\approx\; 5\,\mu{\rm eV}.$$

$$\boxed{T_{\max} \;\sim\; W/k_B \;\sim\; 50\,{\rm mK}\text{–}0.1\,{\rm K}.} \tag{23}$$

Sub-kelvin temperatures (dilution-fridge regime).

End of solutions.